

Paper ID: 1571194120

Title: Benchmarking Transformers and Baselines for Multi-Horizon Stock Return Prediction with Technical and Earnings Features

Authors: Nelson Siu, Jonathan H. Chan

Rebuttal Letter

Dear ICITEE 2025 Program Chairs and Reviewers,

We thank the reviewers for their constructive feedback. We provide below a point-by-point response. All changes are highlighted in the revised manuscript: underlines denote NEW content; *italics* denote reworded/simplified passages.

Reviewer 1

- **Comment.** Limited scope (Dow 30, daily) reduces generalizability; consider broader universes and higher frequency.

Response. We agree and clarified scope boundaries and suggested robustness extensions without adding new experiments. We emphasize cross-universe and higher-frequency tests as future work and add practical guidance on when findings likely transfer.

Where. *Data & Experimental Design*, “Robustness and additional validation” (para 4, lines 180–220, *italicized*); Discussion, “Limitations and threats to validity” (para 3, lines 780–835, new); *Conclusion* (final paragraph, lines 980–1020, *clarified*); *Future Work* (bullets (i)–(ii), lines 1030–1065).

- **Comment.** TFT capacity/fairness: reduced layers/heads may understate transformer potential; request higher-capacity, properly regularized runs.

Response. We acknowledge the constrained-capacity TFT and now make the conditional interpretation explicit. We did not add new runs; instead we added capacity/regularization caveats and clarified that stronger/pretrained transformers may fare better with richer data.

Where. Models, “Transformer model” – “Capacity & regularization” (two new paragraphs, lines 520–575); Fig. ?? caption (new takeaway about compact config, lines 600–615); Discussion, Limitations (para 2, lines 800–820).

- **Comment.** Economic significance: add trading/backtesting perspective (Sharpe, costs) beyond RMSE/DA.

Response. We added an Economic Significance subsection stating no new backtests were run and framing statistical results as non-economic by default; we outline cost-aware evaluation as future work.

Where. Discussion, §Economic Significance (lines 740–780, new).

- **Comment.** Clarity/captions and recent references: expand key figure/table captions; include up-to-date multimodal/transformer citations.

Response. We expanded takeaways in the captions of Table I, Fig. 1 (DA by horizon), and Fig. 2 (regime heatmap), and added 2023–2024 references (PatchTST, TimesFM; BloombergGPT, FinGPT) with contextualized remarks.

Where. Table I caption (lines 655–690, *reworded with new takeaway*); Fig. 1 caption (lines 640–655, *reworded with new summary*); Fig. 2 caption (lines 900–930, *reworded with new*

context); Related Work, final paragraph “Recent transformer & multimodal trends” (lines 320–365, new).

Reviewer 2

- **Comment.** Bridge the stats→economics gap; caution against direct deployment without cost-aware analysis.

Response. Added an explicit caveat that statistical accuracy does not imply profitability and listed key economic metrics and frictions; no new experiments were run.

Where. Discussion, §Economic Significance (lines 740–780, new).

- **Comment.** Sensitivity to extreme/rare regimes (crashes, illiquidity) not explicitly addressed.

Response. We added a tail-event caveat and noted that performance during extreme shocks warrants dedicated evaluation in future work.

Where. Discussion, “Limitations and threats to validity” (lines 800–835, new); *Future Work* (lines 1030–1065, *clarified*).

- **Comment.** Generalization beyond Dow 30 should be stressed.

Response. We strengthened the statement that results may not transfer to small-caps, international markets, or higher-frequency settings and pointed to broader universes as next steps.

Where. *Data & Experimental Design, “Robustness and additional validation”* (para 4, lines 180–220, *clarified*); Discussion, Limitations (lines 800–835, new).

Reviewer 3

- **Comment.** Small feature set; consider alternative data (news, sentiment, macro) and multimodal fusion.

Response. We clarified the intentional restriction to structured technical+earnings features to isolate modeling effects and positioned multimodal extensions as future work; we also added recent multimodal citations.

Where. *Data & Experimental Design, “Features”* (para 7, lines 240–310, *clarified*); Related Work, final paragraph (lines 320–365, new); *Future Work* (bullet (i), lines 1030–1065).

- **Comment.** Compute/efficiency comparison is useful when contrasting heavy TFT vs. simpler baselines.

Response. We added a concise compute/efficiency subsection with parameter scales and training settings to contextualize practical costs; no new timing benchmarks are claimed.

Where. Discussion, §Compute & Efficiency (lines 780–800, new).

Minor edits

Typos, consistency, and formatting refinements throughout (*clarified*); expanded figure/table caption takeaways; added up-to-date references in *Related Work*.

We appreciate the reviewers’ helpful suggestions. We believe these revisions improve clarity, completeness, and the positioning of the contribution. We hope the changes satisfactorily address the minor revisions requested.